

following function:

$$\pi(x) = \text{number of primes less than or equal to } x.$$

The erratic growth of the function $\pi(x)$ gives little hope of finding a simple formula for it. Instead, one is led to study the asymptotic behavior of $\pi(x)$ as x becomes large. About 60 years after Euler's discovery, Legendre and Gauss observed after numerous calculations that it was likely that

$$(1) \quad \pi(x) \sim \frac{x}{\log x} \quad \text{as } x \rightarrow \infty.$$

(The asymptotic relation $f(x) \sim g(x)$ as $x \rightarrow \infty$ means that $f(x)/g(x) \rightarrow 1$ as $x \rightarrow \infty$.) Another 60 years later, shortly before Riemann's work, Tchebychev proved by elementary methods (and in particular, without the zeta function) the weaker result that

$$(2) \quad \pi(x) \approx \frac{x}{\log x} \quad \text{as } x \rightarrow \infty.$$

Here, by definition, the symbol $\approx$ means that there are positive constants $A < B$ such that

$$A \frac{x}{\log x} \leq \pi(x) \leq B \frac{x}{\log x}$$

for all sufficiently large x .

In 1896, about 40 years after Tchebychev's result, Hadamard and de la Vallée Poussin gave a proof of the validity of the relation (1). Their result is known as the prime number theorem. The original proofs of this theorem, as well as the one we give below, use complex analysis. We should remark that since then other proofs have been found, some depending on complex analysis, and others more elementary in nature.

At the heart of the proof of the prime number theorem that we give below lies the fact that $\zeta(s)$ does not vanish on the line $\text{Re}(s) = 1$. In fact, it can be shown that these two propositions are equivalent.

1 Zeros of the zeta function

We have seen in Theorem 1.10, Chapter 8 in Book I, Euler's identity, which states that for $\text{Re}(s) > 1$ the zeta function can be expressed as an infinite product

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

For the sake of completeness we provide a proof of the above identity. The key observation is that $1/(1 - p^{-s})$ can be written as a convergent (geometric) power series

$$1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots + \frac{1}{p^{Ms}} + \cdots,$$

and taking formally the product of these series over all primes p , yields the desired result. A precise argument goes as follows.

Suppose M and N are positive integers with $M > N$. Observe now that, by the fundamental theorem of arithmetic,¹ any positive integer $n \leq N$ can be written uniquely as a product of primes, and that each prime that occurs in the product must be less than or equal to N and repeated less than M times. Therefore

$$\begin{aligned} \sum_{n=1}^N \frac{1}{n^s} &\leq \prod_{p \leq N} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots + \frac{1}{p^{Ms}} \right) \\ &\leq \prod_{p \leq N} \left(\frac{1}{1 - p^{-s}} \right) \\ &\leq \prod_p \left(\frac{1}{1 - p^{-s}} \right). \end{aligned}$$

Letting N tend to infinity in the series now yields

$$\sum_{n=1}^{\infty} \frac{1}{n^s} \leq \prod_p \left(\frac{1}{1 - p^{-s}} \right).$$

For the reverse inequality, we argue as follows. Again, by the fundamental theorem of arithmetic, we find that

$$\prod_{p \leq N} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots + \frac{1}{p^{Ms}} \right) \leq \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Letting M tend to infinity gives

$$\prod_{p \leq N} \left(\frac{1}{1 - p^{-s}} \right) \leq \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

¹A proof of this elementary (but essential) fact is given in the first section of Chapter 8 in Book I.

Hence

$$\prod_p \left(\frac{1}{1 - p^{-s}} \right) \leq \sum_{n=1}^{\infty} \frac{1}{n^s},$$

and the proof of the product formula for ζ is complete.

From the product formula we see, by Proposition 3.1 in Chapter 5, that $\zeta(s)$ does not vanish when $\operatorname{Re}(s) > 1$.

To obtain further information about the location of the zeros of ζ , we use the functional equation that provided the analytic continuation of ζ . We may write the fundamental relation $\xi(s) = \xi(1-s)$ in the form

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \pi^{-(1-s)/2} \Gamma((1-s)/2) \zeta(1-s),$$

and therefore

$$\zeta(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)} \zeta(1-s).$$

Now observe that for $\operatorname{Re}(s) < 0$ the following are true:

- (i) $\zeta(1-s)$ has no zeros because $\operatorname{Re}(1-s) > 1$.
- (ii) $\Gamma((1-s)/2)$ is zero free.
- (iii) $1/\Gamma(s/2)$ has zeros at $s = -2, -4, -6, \dots$.

Therefore, the only zeros of ζ in $\operatorname{Re}(s) < 0$ are located at the negative even integers $-2, -4, -6, \dots$

This proves the following theorem.

Theorem 1.1 *The only zeros of ζ outside the strip $0 \leq \operatorname{Re}(s) \leq 1$ are at the negative even integers, $-2, -4, -6, \dots$*

The region that remains to be studied is called the **critical strip**, $0 \leq \operatorname{Re}(s) \leq 1$. A key fact in the proof of the prime number theorem is that ζ has no zeros on the line $\operatorname{Re}(s) = 1$. As a simple consequence of this fact and the functional equation, it follows that ζ has no zeros on the line $\operatorname{Re}(s) = 0$.

In the seminal paper where Riemann introduced the analytic continuation of the ζ function and proved its functional equation, he applied these insights to the theory of prime numbers, and wrote down “explicit” formulas for determining the distribution of primes. While he did not succeed in fully proving and exploiting his assertions, he did initiate many important new ideas. His analysis led him to believe the truth of what has since been called the **Riemann hypothesis**:

The zeros of $\zeta(s)$ in the critical strip lie on the line $\operatorname{Re}(s) = 1/2$.

He said about this: “It would certainly be desirable to have a rigorous demonstration of this proposition; nevertheless I have for the moment set this aside, after several quick but unsuccessful attempts, because it seemed unneeded for the immediate goal of my study.” Although much of the theory and numerical results point to the validity of this hypothesis, a proof or a counter-example remains to be discovered. The Riemann hypothesis is today one of mathematics’ most famous unresolved problems.

In particular, it is for this reason that the zeros of ζ located outside the critical strip are sometimes called the **trivial zeros** of the zeta function. See also Exercise 5 for an argument proving that ζ has no zeros on the real segment, $0 \leq \sigma \leq 1$, where $s = \sigma + it$.

In the rest of this section we shall restrict ourselves to proving the following theorem, together with related estimates on ζ , which we shall use in the proof of the prime number theorem.

Theorem 1.2 *The zeta function has no zeros on the line $\operatorname{Re}(s) = 1$.*

Of course, since we know that ζ has a pole at $s = 1$, there are no zeros in a neighborhood of this point, but what we need is the deeper property that

$$\zeta(1 + it) \neq 0 \quad \text{for all } t \in \mathbb{R}.$$

The next sequence of lemmas gathers the necessary ingredients for the proof of Theorem 1.2.

Lemma 1.3 *If $\operatorname{Re}(s) > 1$, then*

$$\log \zeta(s) = \sum_{p,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n n^{-s}$$

for some $c_n \geq 0$.

Proof. Suppose first that $s > 1$. Taking the logarithm of the Euler product formula, and using the power series expansion for the logarithm

$$\log \left(\frac{1}{1-x} \right) = \sum_{m=1}^{\infty} \frac{x^m}{m},$$

which holds for $0 \leq x < 1$, we find that

$$\log \zeta(s) = \log \prod_p \frac{1}{1 - p^{-s}} = \sum_p \log \left(\frac{1}{1 - p^{-s}} \right) = \sum_{p,m} \frac{p^{-ms}}{m}.$$

Since the double sum converges absolutely, we need not specify the order of summation. See the Note at the end of this chapter. The formula then holds for all $\operatorname{Re}(s) > 1$ by analytic continuation. Note that, by Theorem 6.2 in Chapter 3, $\log \zeta(s)$ is well defined in the simply connected half-plane $\operatorname{Re}(s) > 1$, since ζ has no zeros there. Finally, it is clear that we have

$$\sum_{p,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n n^{-s},$$

where $c_n = 1/m$ if $n = p^m$ and $c_n = 0$ otherwise.

The proof of the theorem we shall give depends on a simple trick that is based on the following inequality.

Lemma 1.4 *If $\theta \in \mathbb{R}$, then $3 + 4 \cos \theta + \cos 2\theta \geq 0$.*

This follows at once from the simple observation

$$3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2.$$

Corollary 1.5 *If $\sigma > 1$ and t is real, then*

$$\log |\zeta^3(\sigma) \zeta^4(\sigma + it) \zeta(\sigma + 2it)| \geq 0.$$

Proof. Let $s = \sigma + it$ and note that

$$\operatorname{Re}(n^{-s}) = \operatorname{Re}(e^{-(\sigma+it)\log n}) = e^{-\sigma \log n} \cos(t \log n) = n^{-\sigma} \cos(t \log n).$$

Therefore,

$$\begin{aligned} & \log |\zeta^3(\sigma) \zeta^4(\sigma + it) \zeta(\sigma + 2it)| \\ &= 3 \log |\zeta(\sigma)| + 4 \log |\zeta(\sigma + it)| + \log |\zeta(\sigma + 2it)| \\ &= 3 \operatorname{Re}[\log \zeta(\sigma)] + 4 \operatorname{Re}[\log \zeta(\sigma + it)] + \operatorname{Re}[\log \zeta(\sigma + 2it)] \\ &= \sum c_n n^{-\sigma} (3 + 4 \cos \theta_n + \cos 2\theta_n), \end{aligned}$$

where $\theta_n = t \log n$. The positivity now follows from Lemma 1.4, and the fact that $c_n \geq 0$.

We can now finish the proof of our theorem.

Proof of Theorem 1.2. Suppose on the contrary that $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$. Since ζ is holomorphic at $1 + it_0$, it must vanish at least to order 1 at this point, hence

$$|\zeta(\sigma + it_0)|^4 \leq C(\sigma - 1)^4 \quad \text{as } \sigma \rightarrow 1,$$

for some constant $C > 0$. Also, we know that $s = 1$ is a simple pole for $\zeta(s)$, so that

$$|\zeta(\sigma)|^3 \leq C'(\sigma - 1)^{-3} \quad \text{as } \sigma \rightarrow 1,$$

for some constant $C' > 0$. Finally, since ζ is holomorphic at the points $\sigma + 2it_0$, the quantity $|\zeta(\sigma + 2it_0)|$ remains bounded as $\sigma \rightarrow 1$. Putting these facts together yields

$$|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \rightarrow 0 \quad \text{as } \sigma \rightarrow 1,$$

which contradicts Corollary 1.5, since the logarithm of real numbers between 0 and 1 is negative. This concludes the proof that ζ is zero free on the real line $\operatorname{Re}(s) = 1$.

1.1 Estimates for $1/\zeta(s)$

The proof of the prime number theorem relies on detailed manipulations of the zeta function near the line $\operatorname{Re}(s) = 1$; the basic object involved is the logarithmic derivative $\zeta'(s)/\zeta(s)$. For this reason, besides the non-vanishing of ζ on the line, we need to know about the growth of ζ' and $1/\zeta$. The former was dealt with in Proposition 2.7 of Chapter 6; we now treat the latter.

The proposition that follows is actually a quantitative version of Theorem 1.2.

Proposition 1.6 *For every $\epsilon > 0$, we have $1/|\zeta(s)| \leq c_\epsilon |t|^\epsilon$ when $s = \sigma + it$, $\sigma \geq 1$, and $|t| \geq 1$.*

Proof. From our previous observations, we clearly have that

$$|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \geq 1, \quad \text{whenever } \sigma \geq 1.$$

Using the estimate for ζ in Proposition 2.7 of Chapter 6, we find that

$$|\zeta^4(\sigma + it)| \geq c|\zeta^{-3}(\sigma)| |t|^{-\epsilon} \geq c'(\sigma - 1)^3 |t|^{-\epsilon},$$